Sports Fanatic

Product Overview

AI-Powered MLB Intelligence Platform

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Executive Summary

Sports Fanatic is an AI-powered MLB intelligence platform that combines comprehensive baseball data with machine learning predictions and AI-generated editorial content. The platform operates on a two-layer business model: a free public SEO layer that drives organic traffic, and a premium subscription layer that monetizes engaged users.

The product is designed to become the go-to destination for MLB fans seeking daily game intelligence, from casual browsers looking up lineups and news to serious fans who want AI-driven scouting reports and matchup analysis for every game.

Value Proposition

Scouting reports, Game Day Magazines, hit and HR predictions - generated by AI for every player, every matchup, every day. No other platform delivers this breadth of AI-generated MLB analysis at this price point.

Product Overview

1. Free Public Layer (SEO & Acquisition)

The free layer serves as both a content destination and an acquisition funnel. Every page is optimized for search engines with structured data, Open Graph tags, and fast server-rendered HTML. Target: Lighthouse SEO score >= 95.

Public pages include:

- Player profiles for every active MLB player (900+ pages)
- Team pages for all 30 MLB teams with rosters and stats
- Daily lineup pages updated automatically before each game
- MLB news aggregation with daily coverage
- Preview scouting reports (truncated to drive subscription conversion)
- Preview Game Day Magazines (limited content as teaser)
- Full XML sitemaps for Google/Bing indexing

2. Premium Subscription Layer (Revenue)

The premium layer gates high-value AI-generated content behind a Stripe subscription. Subscribers get full, untruncated access to all AI-generated analysis plus real-time features via WebSocket connections.

Premium features include:

- Full AI Scouting Reports - strengths, weaknesses, injury risk, confidence scores, per-player hit/HR probabilities

- Game Day Magazines - comprehensive matchup previews with pitcher analysis, key storylines, weather impact, and AI predictions
- Real-time Dashboard - live in-game updates via WebSocket (Phoenix LiveView)
- Archive Access - browse historical scouting reports and Game Day Magazines
- Priority support and early access to new features (Yearly plan)

Pricing & Subscription Tiers

All plans include a 7-day free trial. Payment is processed through Stripe with full webhook integration for automated provisioning and lifecycle management.

Monthly Plan - $9.99/month

7-day free trial included. Full access to all premium content: scouting reports, Game Day Magazines, real-time predictions, and archive access. Cancel anytime.

Yearly Plan - $79.99/year (Save 33%)

7-day free trial included. Everything in Monthly plus priority support and early access to new features. Best value for committed fans. Equivalent to $6.67/month.

Revenue Model

The freemium model leverages SEO traffic as a zero-cost acquisition channel. Public preview pages (scouting reports and Game Day Magazines) act as conversion funnels with prominent CTAs. Every public preview includes a "Subscribe for Full Access" prompt with trial messaging.

- Acquisition: Organic search (SEO layer drives traffic at zero CAC)
- Activation: Free trial removes friction from first conversion
- Revenue: Monthly recurring revenue (MRR) via Stripe subscriptions
- Retention: Daily fresh content (new games, new reports) creates habitual usage

Key Features - Deep Dive

AI Scouting Reports

Generated daily for every player in a confirmed lineup. Each report includes an overall letter grade, a narrative summary, strengths and weaknesses, injury risk assessment, and hit/HR probability predictions. Reports are generated by Claude AI using current season stats, recent performance trends, and matchup-specific data.

Content Source: Claude AI via Python microservice

Update Frequency: Daily, for all players in confirmed lineups

Free Preview: Overall grade, summary, and top 2 strengths only

Premium Content: Full report with all sections, probabilities, and modifiers

Game Day Magazines

A magazine-style preview generated for every MLB game. Each magazine covers the full matchup: probable pitchers, key storylines, per-batter analysis, weather impact, historical context, and AI-powered outcome predictions. Think of it as a personalized sports magazine delivered fresh for every game, every day.

Content Source: Claude AI + Monte Carlo simulation + ML models

Update Frequency: Daily, one per scheduled game

Free Preview: Headline, matchup preview (500 chars), top 3 storylines

Premium Content: Full analysis, per-batter probabilities, weather, historical data

Hit & HR Predictions

Machine learning models (Gradient Boosting) combined with Monte Carlo simulation (10,000 trials per prediction) generate probability estimates for hits and home runs. These predictions are integrated into scouting reports and Game Day Magazines, giving subscribers a quantitative edge.

ML Model: Gradient Boosting (scikit-learn)

Simulation: Monte Carlo, 10,000 trials, NumPy-vectorized

Inputs: Player stats, pitcher matchup, venue park factors, weather

Real-Time Dashboard

Premium subscribers access a live dashboard powered by Phoenix LiveView (WebSocket). This enables real-time in-game updates without page refreshes - scores, lineup changes, and prediction adjustments stream directly to the browser.

Content & SEO Strategy

The platform generates thousands of unique, crawlable pages automatically. Every page is optimized with structured data (JSON-LD), Open Graph tags for social sharing, Twitter Cards, canonical URLs, and auto-generated XML sitemaps.

Page Inventory (Estimated)

Player Pages: 900+ (one per active MLB player)

Team Pages: 30 (one per MLB team)

Daily Lineup Pages: 1 per game day (~2,430/season)

News Pages: Multiple per day (aggregated)

Scouting Report Pages: 900+ per day (one per player in lineup)

Game Day Magazine Pages: ~15 per game day (~2,430/season)

Total Indexed Pages: Tens of thousands per season, growing daily

SEO Infrastructure

- Server-side rendered HTML (not client-side JS) for full crawlability
- JSON-LD structured data on every page (SportsScoringEvent, Person, Organization)
- Auto-generated XML sitemaps: players, teams, lineups, news, scouting, game day
- Canonical URLs prevent duplicate content issues
- Open Graph + Twitter Card tags for social sharing previews
- Fast response times via Valkey (Redis-compatible) caching layer
- robots.txt configured for search engine access

Conversion Funnel

Free pages are designed as top-of-funnel content that naturally leads to premium conversion. The funnel works as follows:

- DISCOVER: User finds a player/team/game page via Google search
- ENGAGE: User browses free content (stats, lineup, news, preview reports)
- TEASE: Scouting report and Game Day previews show partial content with upgrade CTA
- CONVERT: User clicks "Start Free Trial" (7 days, no charge)
- RETAIN: Daily fresh AI content creates habitual return visits

Data Pipeline & Automation

The platform runs a fully automated content pipeline. No manual editorial work is required for daily operations. Nine background workers handle data ingestion, AI content generation, cache warming, and housekeeping on automated schedules.

Automated Daily Operations

- Fetch MLB schedule and game data from MLB Stats API
- Fetch confirmed lineups for today's games
- Aggregate and ingest MLB news
- Generate AI scouting reports for all players in confirmed lineups
- Generate Game Day Magazines for all scheduled games
- Run ML prediction models (hit/HR probabilities)
- Warm cache for high-traffic pages (reduces latency)
- Regenerate XML sitemaps for search engines
- Clean up expired data and stale cache entries

External Data Sources

MLB Stats API: Official game data, schedules, lineups, player stats

Claude AI (Anthropic): Scouting report and Game Day Magazine generation

Exa AI Search: Contextual web search for scouting research

FantasyNerds: Supplemental player and fantasy data

Content Freshness

Caching ensures fast page loads while content stays fresh. Cache durations are tuned per content type:

Today's lineups: 2-minute cache (near real-time)

Player pages: 5-minute cache

News pages: 5-minute cache

Game Day Magazines: 10-minute cache

Team pages: 10-minute cache

Scouting reports: 15-minute cache

Past lineups: 24-hour cache (historical, rarely changes)

Sitemaps: 6-hour cache

Competitive Positioning

Sports Fanatic occupies a unique position at the intersection of AI-generated content and MLB data. Below is how the platform differentiates from existing alternatives.

vs. Traditional Sports Media (ESPN, The Athletic, etc.)

- AI generates content for EVERY player and EVERY game - not just marquee matchups
- Coverage scales automatically; no editorial bottleneck
- Quantitative predictions (ML + Monte Carlo) complement narrative analysis
- Significantly lower content production cost per article

vs. Stats/Data Sites (Baseball Reference, FanGraphs)

- AI translates raw data into readable narrative scouting reports
- Game Day Magazine format is accessible to casual fans, not just analysts
- Predictive (forward-looking probabilities) vs. purely historical stats

vs. Fantasy/Betting Platforms (DraftKings, FanDuel)

- Independent analysis not tied to betting/fantasy platform incentives
- Deeper per-player narrative analysis (scouting reports vs. just projections)
- Magazine-style content creates engagement beyond transaction moments

Key Differentiator

No other platform generates AI-powered scouting reports and magazine-style matchup previews for every MLB game, every day, at a $9.99/month price point. The combination of breadth (every game), depth (full scouting + predictions), and accessibility (narrative format) is unique in the market.

Key Metrics & KPIs

Acquisition & Traffic

- Organic search impressions and clicks (Google Search Console)
- Unique visitors per day/week/month
- Pages per session (engagement depth)
- Top landing pages by traffic volume
- Lighthouse SEO score (target: >= 95)

Conversion

- Free-to-trial conversion rate (visitor -> trial signup)
- Trial-to-paid conversion rate (trial -> paying subscriber)
- Conversion by entry page (which content converts best)
- CTA click-through rate on preview pages

Revenue

- Monthly Recurring Revenue (MRR)
- Annual Recurring Revenue (ARR)
- Average Revenue Per User (ARPU)
- Monthly vs. Yearly plan mix
- Customer Acquisition Cost (CAC) - expected near-zero via SEO

Retention & Engagement

- Monthly churn rate
- Daily Active Users (DAU) among subscribers
- Content consumption: reports viewed per subscriber per day
- Return visit frequency
- Net Promoter Score (NPS)

Content & Operations

- Scouting reports generated per day
- Game Day Magazines generated per day
- AI content generation success rate (pipeline health)
- Average page load time
- Cache hit rate

Technology Summary (Non-Technical)

This section provides a high-level overview of the technology stack for stakeholders who need to understand capabilities and constraints without engineering detail.

Platform Architecture

- Web application built on Elixir/Phoenix - a modern, high-performance framework known for handling real-time features and high concurrency
- AI microservice (Python) handles machine learning predictions and Claude AI content generation
- PostgreSQL database stores all structured data (players, teams, games, content, users)
- Valkey cache layer ensures fast page loads even under high traffic
- Stripe handles all payment processing and subscription lifecycle
- Docker containers enable consistent deployment across environments

AI & Machine Learning

- Claude AI (Anthropic) generates narrative scouting reports and Game Day Magazines
- Gradient Boosting ML models predict hit and home run probabilities
- Monte Carlo simulation (10,000 trials) adds statistical rigor to predictions
- Exa AI web search provides real-time context for scouting research

Scalability Notes

- All content generation is automated - no editorial staff required for daily operations
- Adding new content types (e.g., weekly roundups, playoff specials) requires engineering work but follows established patterns
- The SEO layer can scale to tens of thousands of pages without performance impact
- Real-time features (LiveView) support thousands of concurrent WebSocket connections

Potential Roadmap Opportunities

The following are product expansion opportunities that the current architecture could support. These are not committed roadmap items but represent natural extensions of the platform's capabilities.

Content Expansion

- Weekly roundup magazines (AI-generated season narrative)
- Playoff and postseason special editions
- Historical "On This Day" content for evergreen SEO
- Fantasy baseball integration and daily recommendations
- Multi-sport expansion (NFL, NBA, NHL) using the same AI pipeline

Monetization

- Tiered premium plans (e.g., Basic, Pro, Enterprise)
- Team or league-specific subscriptions at lower price points
- API access tier for data consumers and app developers
- Affiliate partnerships with sports betting platforms
- Sponsored content integration (clearly labeled)

Engagement & Retention

- Push notifications for game-day content and prediction results
- Email digest (daily/weekly) for subscribers
- Personalized "My Teams" dashboard based on user preferences
- Social sharing features (shareable prediction cards)
- Community features (comments, prediction leagues)

Platform

- Native mobile app (iOS/Android)
- Widget/embed program for partner sites
- Alexa/Google Assistant voice integration for daily predictions

Glossary

Scouting Report

An AI-generated analysis of an individual MLB player covering strengths, weaknesses, injury risk, and probability predictions for upcoming games.

Game Day Magazine

An AI-generated magazine-style preview of an MLB game covering both teams, pitchers, key storylines, weather impact, and outcome predictions.

SEO (Search Engine Optimization)

The practice of optimizing web pages to rank higher in Google and other search engines, driving free organic traffic.

JSON-LD

A structured data format embedded in web pages that helps search engines understand content (e.g., identifying a page as being about a specific athlete).

Freemium

A business model where basic content is free and premium features require a paid subscription.

MRR (Monthly Recurring Revenue)

The predictable monthly revenue from active subscriptions.

CAC (Customer Acquisition Cost)

The cost to acquire a new paying customer. SEO-driven acquisition targets near-zero CAC.

Churn Rate

The percentage of subscribers who cancel in a given period.

Monte Carlo Simulation

A statistical technique that runs thousands of randomized trials to estimate probabilities. Used here to predict hit and home run likelihood.

LiveView

A real-time web technology that pushes updates to the browser without page refreshes, used for the premium dashboard.

Valkey

A high-speed data cache (compatible with Redis) that stores frequently accessed pages for fast delivery.

Stripe

Third-party payment processing platform that handles subscriptions, trials, and billing.

Oban

Background job processing system that automates scheduled tasks like data fetching and content generation.